RAJALAKSHMI ENGINEERING COLLEGE

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T2

Started on	Friday, 21 August 2026, 1:55 PM
State	Finished
Completed on	Friday, 21 August 2026, 1:57 PM
Time taken	1 min 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n, count = 0;
    if (scanf("%d", &n) != 1) return 0;

    count++; // Outer loop initialization (i = n / 2)

    for (int i = n / 2; i < n; i++) {
        for (int j = 1; j < n; j = 2 * j) {
            for (int k = 1; k < n; k = k * 2) {
                count++; // inner loop comparison check
                count++; // c++ execution inside inner loop
            }
            count++; // inner loop condition check that fails
            count++; // middle loop comparison check
        }
        count++; // middle loop condition check that fails
        count++; // outer loop comparison check
    }
    count++; // outer loop condition check that fails

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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